

NIHARIKA VADLAMUDI

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EDUCATION

International Institute of Information Technology, Hyderabad

B.Tech & MS by Research (Part Time), ECE

August 2018 - Feb 2025

GPA: 8.00

Thesis : High Precision Text Line Segmentation in Palm Leaf Manuscripts.

PUBLICATIONS

[ICDAR'23 · Accepted] (*First Author*) SeamFormer: High Precision Text Line Segmentation for Handwritten Documents

[ICPR'24 · Accepted] (*Second Author*) LineTR: Towards Universal Text Line Segmentation for Handwritten Documents

[ML4H 2025 · Accepted] (*Co-Author*) Rad-Phi4-Vision-CXR: A Compact Multimodal Assistant for Versatile Radiology Workflows

[MLHC 2026 · Accepted] (*First Author, Equal Contribution*) CARE-X: Towards Clinically Useful Radiology VLMs with Auxiliary Supervision, Reward-Aligned Learning, and Tool-Augmented Measurement

[EACTS 2026 · Accepted] (*Co-Author*) Early Detection of Borderline Aortic Dilatation on Chest Radiographs Using a Multimodal Morphometric Agent

[AHA 2026 · Submitted] (*First Author*) Automated Multi-Landmark Morphometric Screening for Ascending Aortic Dilatation on Chest Radiography: A Proof-of-Concept Study

INDUSTRIAL EXPERIENCE

Microsoft Research

Research Intern

October 2025 - June 2026

CARE Team , Bengaluru

- **Led coordination with Narayana Health hospitals**, managing dataset access and driving model evaluations across incremental versions against clinically defined exit criteria spanning multiple pathologies.
- **Developed an optimized** multimodal VLM for rare and critical ICU pathologies , leveraging a **discriminative loss** to reinforce cross-entropy loss within a VLM setting.
- **Led tube detection and abnormality placement** efforts – including dataset creation, model training, and evaluation – for classifying tube placement and associated abnormalities.
- **Developed an aortic dilatation detection pipeline**, independently collaborating with the NH team on a **novel geometric approach** for detecting ascending aortic enlargement (AAE).
- **Conducted a systematic literature review** of radiology foundation and baseline models, collating published benchmark results for comparative analysis.

Microsoft

Software Developer I

August 2022 – August 2025

Torus Buildout, Hyderabad — Experiences + Devices

- Automated and optimized cloud orchestration workflows, enabling two critical role bring-ups and reducing buildout time by approximately **5%**.
- Developed and deployed a configurable certificate provisioning workflow in **CredSMART**, adding support for publishing **public trusted root** and **public certificates** to Azure Active Directory.
- Integrated a core service - **Torus SecretsBootstrapper** service with two core services for end-to-end enablement of secret management in upcoming sovereign cloud pipelines.
- Conducted integration testing for 30+ **CredSMART APIs**, supporting 100+ partner teams in secure service key and certificate workflows.
- Automated core security services (**Lockbox, IDM**) using the Buildout Coordinator Tool, improving reliability and minimizing manual effort.
- Implemented the **Torus Configuration Syncer Service**, enabling dynamic, code-free configuration management across multiple datacenter services.

Microsoft

Software Intern

May, 2021 - July, 2021

IC3 Admin Scenarios Team ,Hyderabad

- Enhanced pagination and search features in **Modern Portal (MoPo)** Admin, improving admin workflow efficiency and user experience.

- Designed and implemented Azure Cosmos DB (NoSQL) APIs to drive performance benchmarking and enable COGS optimization for custom Azure server configurations.
- Developed a scalable data ingestion tool handling millions of records, leading to **multi-million-dollar cost savings** across cloud operations.

Microsoft

June 2020 – July 2020

Codess Mentee

Remote

Selected for the **Microsoft Codess Program** (2020), a competitive initiative to empower women in software engineering. Designed and implemented a mini Mars Rover navigation system using core AI path planning algorithms (**A***, **Dijkstra's**, **DFS**, **Bidirectional BFS**) with a focus on ethical AI principles. **Offered a direct internship opportunity** for Summer 2021 based on performance.

RESEARCH EXPERIENCE

Center For Visual Information Technology (CVIT)

May, 2020 - August, 2023

Researcher

IIIT Hyderabad

- As a member of the **IHDIA** research group, collaborated with **Dr. Ravi Kiran S** to investigate interdisciplinary methodologies for analyzing text layouts within historical palm leaf manuscripts. This collaboration culminated in the completion of my Master's thesis and **two publications**.
- Implemented the SeamFormer model into the *Bhashini* platform to facilitate the processing of ancient Indic manuscripts, providing support to a diverse community of over **1000+** users and academicians.

Vision Lab

December 2023 - June 2024

Research Engineer

IIIT Delhi

- Collaborated with **Dr. Saket Anand** on a government-funded project titled '*Automated Road Inspection using Videos for E-Marg*'.
- Key contributions include fine-tuning state-of-the-art vision models such as DeepLabV3+, SAM, and EfficientNet for small scale dataset. Developed an integrated model exhibiting a **6%** increase in accuracy for Road Quality Assessment.
- Researching to design novel Active Learning (AL) methods and frame selection strategies for video modalities to significantly reduce annotation costs for large-scale raw data.

IHub-Data

June 2022 - August 2022

ML Research Fellow

IIIT Hyderabad

- Worked with **Dr. UD Priyakumar** in the domain of computational drug design for **de novo drug molecular generation**.
- Extended state-of-the-art baseline models for multi-property prediction on the large scale Harvard Therapeutics Data Commons (TDC) dataset - ADMET.
- Contributed to designing multiple GNN encoder-decoder architectures utilizing Graph Isomorphism Networks (GINs), Graph Attention Networks (GATs), Graph Convolutional Networks (GCNs), and other related methodologies, while also conducting literature reviews.

ACHIEVEMENTS

- **Microsoft Global Hackathon 2024**: Developed a certificate troubleshooting tool and was placed **3rd** among 20+ projects in the within our organization.
- **Merit List**: Awarded Merit List recognition for the Monsoon 2021 semester at IIIT Hyderabad.
- **Olympiads**: Recognized with an excellence certificate for ranking in the **top 10%** of participants statewide in the Biology Olympiad organised by the Indian Association of Physics Teachers (IAPT).

SKILLS

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| • C++ / C / C# | • PyTorch |
| • .NET Framework | • TensorFlow |
| • MATLAB | • Python (OpenCV, Scikit-Learn, NumPy, etc.) |
| • LaTeX | • Weights & Biases (WandB) |